Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 22

Direct Proof

Covers Chartrand & Zhang, Section 3.2

Status of this set

Ungraded practice. Solutions released with it. Write every proof in complete sentences — that is the whole exercise.

Part I: How to Practice This Material

Four steps, every time

The template

1. **Assume the hypothesis** and write it as an equation using the definition. "Let n be odd. Then $n = 2k + 1$ for some integer k."
2. **Substitute** into the expression you must analyse.
3. **Algebra** until the target form appears.
4. **Factor out the 2**, say the remaining factor is an integer, and conclude.

Step 4 is where the proof finishes. Everything before is arithmetic.

The definitions, exactly

n is **even** if $n = 2a$ for some *integer* a.
n is **odd** if $n = 2b + 1$ for some *integer* b.
You may assume that sums, products, and negatives of integers are integers. You may *not* assume "even plus even is even" — that is a theorem, not an axiom.

Three mistakes to expect

1. **Omitting "for some integer k."** The last line depends on it.
2. **Stopping at $2m - 1$.** Odd means $2b + 1$. Adjust the algebra.
3. **Using the same letter for two independent quantities.** If $m = 2a$ and $n = 2a$, you have assumed $m = n$.

Part II: Practice Problems

Section A — Parity

1. Prove: if n is an odd integer, then $7n + 4$ is odd.

2. Prove: if n is an even integer, then $3n^2 + n$ is even.

3. Prove: if n is an odd integer, then $n^2 + 3$ is even.

Section B — Two variables

4. Prove: if m and n are both odd integers, then $m + n$ is even.

5. Prove: if m is even and n is odd, then mn is even.

Section C — Beyond parity

6. Prove: if r and s are rational numbers, then rs is rational.

Section D — Writing

7. In five to eight complete sentences, explain why a direct proof of $P \Rightarrow Q$ only needs to consider the case where P is true. Refer to the truth table for the implication.

Section E — Challenge (optional)

8. Prove: if n is an odd integer, then n^3 is odd. Then say what your proof shows about n^4 when n is odd, without a new computation.